# Lecture 6.2: Structural Models

AREC 570 / ECON 530 · Unit 6

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# Before class

### Assigned

- Nevo and Whinston, “Taking the Dogma out of Econometrics: Structural Modeling and Credible Inference” (Nevo and Whinston 2010)
- Nevo, “A Practitioner’s Guide to Estimation of Random-Coefficients Logit Models of Demand” (optional) (Nevo 2000a)

Readings and course handouts are available on Canvas.

### Also used in these slides

- Woerman, ResEcon 703 Week 1 slides, the source of this lecture’s structure (Woerman 2022)
- Example paper: Nevo, mergers in ready-to-eat cereal (Nevo 2000b)
- Timmins and Schlenker, from Lecture 6.1 (Timmins and Schlenker 2009)
- For more detail: the original version of these slides

# Structural models in a nutshell

**If you take away one idea today:** A structural model describes how people or firms make choices, estimates the parameters of that behavior from data, and uses the estimated model to study a change

1. State who chooses what and what shapes those choices
2. Use observed choices to estimate parameters such as price sensitivity
3. Change the policy or market setting in the model
4. Solve the model again to predict choices, prices, or welfare under that change

For a proposed cereal merger, this means estimating how shoppers substitute between brands, then predicting prices under new ownership

Sources: Nevo and Whinston (2010), pages 73 to 77; Nevo (2000b), pages 395 to 397.

# Overview

- An economic research design (a source of variation and an identifying assumption) is meant to answer causal questions
- Some questions cannot be answered that way:
  - the change has not happened yet
  - the answer has to be in dollars of welfare
  - you need to know *why* an outcome changed
- For these, papers write down an economic model and estimate it: a **structural model**

# What is structural econometrics?

# One question answered two ways: how does a merger change prices?

**Reduced form.** Compare prices of the merging firms' products with prices of other products, before and after a merger that already happened. The estimate is the price effect of that merger.

**Structural.** Estimate how consumers substitute between products and how firms set prices. Then solve for prices with the new ownership. This works for a merger that has not happened, and it gives a welfare measure.

Our example: in 1992, Post bought Nabisco's cereals, including Shredded Wheat. New York sued to stop it.

Sources: Nevo and Whinston (2010), pages 73 to 77; Nevo (2000b), page 396.

# A reduced-form model: difference in differences for the Post-Nabisco merger

$$\ln p_{jct} = \beta_1 \mathbf{1}\{\text{Post}\}_j \times \mathbf{1}\{\text{Post-merger}\}_t + \beta_2 \mathbf{1}\{\text{Nabisco}\}_j \times \mathbf{1}\{\text{Post-merger}\}_t + \beta_3 \mathbf{1}\{\text{Post-merger}\}_t + \theta_{jc} + \tau t + \varepsilon_{jct}$$

- $p_{jct}$: price of cereal brand $j$ in city $c$ in quarter $t$
- **Coefficients of interest:** $\beta_1$ and $\beta_2$, the change in Post and Nabisco prices after the merger, relative to other brands
- $\beta_3$: the change for all brands; $\theta_{jc}$: brand-by-city fixed effects; $\tau t$: a time trend
- **Identifying assumption:** without the merger, Post and Nabisco prices would have followed other brands

No economic model is needed. The comparison brands supply the counterfactual.

Our construction for this lecture; the design follows Miller and Weinberg (2017).

# A structural model writes down behavior, then estimates its parameters

1. **Economic theory:** who chooses what, and what they maximize. Consumers pick the cereal they like best; firms set prices to maximize profit
2. **Unobservables:** what the researcher cannot see, such as product quality known to consumers and firms
3. **Estimation:** recover the parameters of behavior, such as how sensitive consumers are to price

Those parameters answer questions the data cannot answer directly: what if two firms merged?

Sources: Woerman (2022), slides 11 and 23 to 24; Timmins and Schlenker (2009), Section 2.

# Why add structure?

# Three reasons to add structure, each from the merger question

| Reason | In the merger question |
|---|---|
| **The change has not happened** | A proposed merger has no post-merger data |
| **The answer must be in dollars** | Consumer welfare needs a model of what consumers value |
| **You need the mechanism** | Did prices rise because brands are close substitutes, or because costs changed? |

A structural model also recovers things no dataset records, such as marginal costs.

Sources: Woerman (2022), slide 16; Timmins and Schlenker (2009), Section 2.1.

# Should you always add structure? It depends on the question and the data

- **Costs of structure:** it needs theory that fits the setting, and assumptions to connect the theory to the data
- If those assumptions are wrong, the answers are wrong, and readers may not see the assumptions
- **Strengths of reduced form:** with a good design, a causal estimate with fewer and more visible assumptions
- **They work together:** a reduced-form estimate can check a model's prediction

Sources: Woerman (2022), slides 17 and 18; Nevo and Whinston (2010), pages 70 to 71.

# How is a structural model estimated?

# Some structural models are estimated with OLS: a production function

A farm produces output $Y$ from capital $K$ and labor $L$ with a Cobb-Douglas technology:

$$Y_t = AK_t^{\alpha} L_t^{\beta} \quad \Rightarrow \quad \ln Y_t = \ln A + \alpha \ln K_t + \beta \ln L_t + \varepsilon_t$$

- $\alpha$ and $\beta$ are parameters of the technology: the percent change in output from a 1% change in each input
- Add an error $\varepsilon_t$ and assume $E[\varepsilon_t \mid K_t, L_t] = 0$; then **OLS** estimates $\alpha$ and $\beta$
- The model is structural because the coefficients come from the theory, even though the estimator is OLS
- **Is the assumption reasonable?** If $\varepsilon_t$ includes productivity the farmer knows when choosing inputs, inputs are endogenous

Source: adapted from Woerman (2022), slide 25.

# Some are estimated with 2SLS: logit demand for cereal

Consumers choose the brand with the highest utility. With the logit assumption, market shares give a linear equation:

$$\ln s_{jt} - \ln s_{0t} = x_{jt}\beta - \alpha p_{jt} + \xi_{jt}$$

- $s_{jt}$: market share of brand $j$ in market $t$; $s_{0t}$: share buying no cereal; $x_{jt}$: characteristics
- $\xi_{jt}$: quality the researcher does not see. Firms see it and price higher, so **price is endogenous**
- Nevo's instrument: the same brand's price in other cities, which shares its cost

| | OLS | 2SLS |
|---|---|---|
| Price coefficient | −8.57 | about −17 |
| Median own-price elasticity | −1.60 | about −3.2 |

Sources: Berry (1994); Nevo (2000b), pages 405 and 418 to 420, Table B1 (columns 1 and 4 to 6).

# When tastes vary across consumers, estimation needs a numerical search

Logit gives everyone the same tastes, so lost sales go to rivals in proportion to their market shares. Random coefficients let tastes vary:

$$\alpha_i = \alpha + \pi \,\text{income}_i + \sigma v_i$$

- Consumer $i$'s price sensitivity = average + a part that varies with income + an unobserved part
- Shares become an average over many consumers, with no formula, so there is no linear regression to run
- **Numerical estimation (GMM):** the computer guesses the parameters, computes predicted shares, compares them with observed shares, and repeats until the fit is best
- This is the BLP model (Berry, Levinsohn, and Pakes, 1995)

Sources: Berry et al. (1995); Nevo (2000a), pages 519 to 533; Woerman (2022), slide 28.

# Example: Nevo (2000), mergers in ready-to-eat cereal

# Nevo's question: what are the price and welfare effects of cereal mergers?

- **Traditional merger analysis** uses market shares to compute a concentration measure (the HHI)
- **Its limit:** shares say nothing about how close two brands are as substitutes
- **Nevo's approach:** estimate demand with random coefficients, recover marginal costs, simulate prices after each merger
- **Five mergers:** two completed (Post and Nabisco; General Mills and Chex), one called off, two hypothetical
- **Data:** supermarket scanner data on 24 brands in 45 cities, 1988 to 1992

Sources: Nevo (2000b), pages 395 to 397 and 404.

# Result 1: Grape Nuts is the closest substitute for Shredded Wheat

| Percent change in the share of … when the price of … rises 1% | Grape Nuts | Shredded Wheat |
|---|---|---|
| Shredded Wheat (Nabisco) | **0.115** | −2.268 |
| Grape Nuts (Post) | −2.096 | **0.070** |
| Raisin Bran (Post) | 0.036 | 0.034 |
| Cheerios (General Mills) | 0.037 | 0.024 |
| Corn Flakes (Kellogg) | 0.000 | 0.001 |

Under logit, every number in a column would be the same.

Source: Nevo (2000b), Table 3 (selected rows and columns; medians over 900 city-quarter markets).

# Result 2: small predicted price increases for Post-Nabisco, close to what happened

| Merger | Largest predicted price increase |
|---|---|
| Post and Nabisco (approved) | Shredded Wheat 3.1%; Grape Nuts 1.5% |
| General Mills and Nabisco (called off) | Shredded Wheat 7.5% |
| General Mills and Chex (approved) | Chex 12.2% |

- A 5% cut in marginal cost would offset the Post-Nabisco increase
- **Check against what happened:** Shredded Wheat's 1995 price was about equal to its 1993 price, close to the prediction with 5% cost savings
- Being the closest substitute does not mean a large price increase: the amount of substitution was small

Sources: Nevo (2000b), pages 411 to 416, Tables 5 and 6.

# Writing up a structural model

# A structural section reports five items

| Item | In Nevo (2000) |
|---|---|
| 1. The model's primitives | Consumer utility, firms' pricing, Bertrand-Nash equilibrium |
| 2. Which variation identifies which parameter | Other cities' prices identify the price coefficient |
| 3. The estimator, in one paragraph | GMM, with details in an appendix |
| 4. How well the model fits or validates | Predictions compared with post-merger prices |
| 5. The counterfactuals it is used for | Five mergers, with welfare |

This is row 10 of the empirical strategy checklist.

# What to think about for your proposal

- **The question:** can a design answer it, or does it need a model? Name the reason
- **The estimator:** OLS, 2SLS, or a numerical method, and why
- **The assumptions:** which ones drive the answer, and what happens if they change?
- **The evidence:** what reduced-form estimate supports the model, or what could check it?

# References


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Woerman, Matt. 2022. *Week 1: Structural Estimation*. Lecture slides, ResEcon 703: Topics in Advanced Econometrics, University of Massachusetts Amherst. https://github.com/woerman/ResEcon703.

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